Complex bashing

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This handout is intended for intermediate complex bashers. You should be able to complete `BGW-complex` or solve most of the problems in Section 3. The main content of this handout is Section 4, which contains 13 walkthroughs in rough order of increasing difficulty. You should try these walkthroughs yourself before reading them.

> [Complex bashing] will keep you alive, even if you are an inch from death, but at a terrible price. You have slain something pure and defenseless [a geometry problem] to save yourself, and you will have but a half-life, a cursed life, from the moment [you write "let (ABC) be the unit circle."]
>
> —*Harry Potter and the Philosopher's Stone*

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§1 Introduction

My style of geometry is to complex bash. Let me show you how. First, here is some advice for doing computations.

- Keep everything factored for as long as possible. When checking if something is real or complex, factor it first.
- Troubleshoot often.
  - Expressions must always be homogeneous (i.e., of the same degree). For example, $bc(a^2\overline{z} + b + c)$ is homogeneous of degree 3, since each term has degree 3. If you ever see something like $bc(a^2\overline{z} + 1)$, something has gone horribly wrong.
  - Plug in $b = c$ or all ones to make sure your expression stays the same from step to step.
  - Any points you construct that are symmetric in B and C better have coordinates that are symmetric in b and c.
- Multiplication with a complex number is a dilation and a rotation, while addition with a complex number is a translation. So you can spam linear functions (compositions of multiplications and additions) to simplify computations.
- Your scratch paper should be in landscape mode.
- Let $f(z) = \frac{z}{\overline{z}}$. Then $f(z) = 1$ if and only if z is real and $f(z) = -1$ if and only if z is pure imaginary. A lot of the time, f is pretty easy to compute, since if a and b are on the unit circle, then $f(a+b) = ab$ and $f(a-b) = -ab$. Also, $f(zw) = f(z)f(w)$. So if a, b, and c are on the unit circle, you can do things like

$$f\left(\frac{(a-b)(a-c)(b+c)}{(a^2+bc)(b-c)}\right) = \frac{(-ab)\cdot(-ac)\cdot bc}{a^2bc\cdot(-bc)} = -1.$$

Again: checking is the MOST important thing when bashing. Slow and steady really does win the race. Many times, I've had to redo a bash because I wasn't careful enough the first time, and I was using the wrong coordinate for a certain point the whole time. Check often. Check thoroughly. Any checking procedure short of doing every computation twice is acceptable. Check more often if you are low on sleep, using a rougher setup, or if (god forbid) you are breaking homogeneity.

§2 Formula sheet

§2.1 Useful formulas

For quick lookup, we include the most important formulas. But you really should already know these.

- For all z lying on the unit circle, $\overline{z} = \frac{1}{z}$.
- The midpoint of segment AB is $\frac{a+b}{2}$.
- The foot from P to $\overline{AB}$ is $\frac{1}{2}(a+b+p-ab\overline{p})$, given that a and b lie on the unit circle.
- The points A, B, and C are collinear if and only if $\frac{a-c}{b-c} \in \mathbb{R}$.
- We have $\measuredangle ABC = \measuredangle DEF$ if and only if $\frac{a-b}{c-b} \div \frac{d-e}{f-e} \in \mathbb{R}$.
- The intersection of lines $\overline{AB}$ and $\overline{CD}$ is $\frac{ab(c+d)-cd(a+b)}{ab-cd}$, given that a, b, c, and d lie on the unit circle. In particular, the intersection of tangents $\overline{AA}$ and $\overline{BB}$ is $\frac{2ab}{a+b}$.
- The points A, B, C, and D are concyclic if and only if $\frac{(a-b)(c-d)}{(a-c)(b-d)} \in \mathbb{R}$.
- The circumcenter of triangle A, B, and C is

$$\begin{vmatrix} 1 & a & a\overline{a} \\ 1 & b & b\overline{b} \\ 1 & c & c\overline{c} \end{vmatrix} \div \begin{vmatrix} 1 & a & \overline{a} \\ 1 & b & \overline{b} \\ 1 & c & \overline{c} \end{vmatrix}.$$

  This matrix formulation is ONLY useful if a, b, and c have nice symmetry, and you can perform row operations to greatly simplify the computation. This case is somewhat rare. It's almost always better to translate one of the points to zero. Then the circumcenter of A, B, and O (where O is the origin) is

$$\frac{ab(\overline{b}-\overline{a})}{a\overline{b}-\overline{a}b} = \frac{\frac{a}{\overline{a}} \cdot b - \frac{b}{\overline{b}} \cdot a}{\frac{a}{\overline{a}} - \frac{b}{\overline{b}}}.$$

  (The reason I included the second, seemingly awkward formulation is that $\frac{a}{\overline{a}}$ and $\frac{b}{\overline{b}}$ often have nice forms.)

- Let lines AX and BY intersect at a point P. The second intersection of (ABP) and (XYP) is the center of the spiral similarity sending $\overline{AB}$ to $\overline{XY}$, which is $\frac{ay-bx}{a+y-b-x}$.

§2.2 Less useful formulas

We also include some less useful formulas.

- Let $\triangle ABC$ have incircle (or A-excircle) contact points D, E, and F. If (DEF) is the unit circle, the center of (ABC) is $\frac{2(d+e+f)def}{(d+e)(e+f)(f+d)}$. The two intersections between (DEF) and (ABC) are the roots of the following two equivalent quadratics.

$$2(de+ef+fd)x^2 - ((d+e)(e+f)(f+d)+4def)x + 2def(d+e+f) = 0$$
$$\overline{a}(2d+a)x^2 - (d^2\overline{a}+2d+a+2da\overline{a})x + da(d\overline{a}+2) = 0.$$

- Let $f, g \in \mathbb{C}[x]$ and let h be an irreducible quadratic with roots r and s. Let $n = \max(\deg f, \deg g) + 1$. Then $\frac{f(r)}{g(r)} = \frac{f(s)}{g(s)}$ if and only if $\alpha f(x) + \beta g(x) + C(x)h(x) = 0$ for some α, β, and C not all zero. Interpreting $\mathbb{C}[x]$ as a vector space, this holds if and only if the determinant of the $n \times n$ matrix with one row as the coefficients of f, one row as the coefficients of g, and $n-2$ rows as the coefficients of $x^i h(x)$ for $0 \le i < n-2$ is zero. This is probably trivial by resultants or something but I don't know resultants.

- Let $ABCD$ be a convex quadrilateral inscribed in the unit circle of the complex plane. It is possible to choose complex numbers a, b, c, and d such that $A = a^2$, $B = b^2$, $C = c^2$, $D = d^2$, and the midpoints of $\widehat{AB}$, $\widehat{BC}$, $\widehat{CD}$, $\widehat{DA}$, $\widehat{ABC}$, $\widehat{BCD}$ are $-ab$, $-bc$, $-cd$, $+da$, $+ac$, and $+bd$.

> **Remark** (A stupid-ass way to remember this)**.** Consider the clockwise arcs. Write the four arcs between adjacent vertices, then the two arcs between opposite vertices. Sort the arcs in each category in *alphabetical order*. The midpoints of the first three arcs have negative sign and the midpoints of the second three arcs have positive sign.

§2.3 Jacobi bialternant formula

This formula is rarely useful, but when it is, it grants the user an incredible amount of aura. **Skip this formula for now if you're less experienced.**

We'll need some theory. A partition $\lambda = \lambda_1\lambda_2\lambda_3\ldots$ is a weakly decreasing sequence of nonnegative integers λ_i with finitely many nonzero elements. We use the French convention that the Young diagram of a partition λ is the subset of $\mathbb{Z}^2_{>0}$ such that (i,j) is in the Young diagram of λ if and only if $i \le \lambda_j$. We will identify λ with its Young diagram, hence we will use λ to refer to both the partition λ and its Young diagram. Let S_n be the symmetric group, so S_n is the set of permutations acting on $\{x_1, \ldots, x_n\}$. For a diagram λ, a tableau is a function τ from λ to any totally ordered set X, usually $\mathbb{Z}_{>0}$. We say a tableau is semistandard if its rows are weakly increasing and its columns are strictly increasing, i.e., if $i' > i$ and $j' > j$, we have $\tau(i,j) \le \tau(i',j)$ and $\tau(i,j) < \tau(i,j')$. The content or weight $\gamma(\tau)$ of a tableau τ is the sequence $(m_1, m_2, \ldots)$, where m_i is the cardinality of the preimage of i with respect to τ. For example, this is a semistandard tableau on $\lambda = 5442$ with content $(1,2,1,1,1,1,2,4,1,1)$.

7	8			
4	6	9	10	
3	5	8	8	
1	2	2	7	8

If λ and μ are two partitions of n, define the Kostka number $K_{\lambda\mu}$ to be the number of semistandard Young tableaux of shape λ and content μ.

Observe that S_n acts on $\mathbb{C}[x_1, \ldots, x_n]$ by permutation of variables, so for a partition λ, define

$$a_\lambda = \sum_{w \in S_n} \operatorname{sgn}(w) w(x_1^{\lambda_1} \cdots x_n^{\lambda_n}) = \det(x_i^{\lambda_j})_{1 \le i,j \le n}.$$

We will now define the monomial symmetric functions. Let λ be a partition of length n and let m_i be the number of times i appears in λ. Then

$$m_\lambda(x_1, \ldots, x_n) := \frac{1}{m_0! \cdots m_n!} \sum_{w \in S_n} w(x_1^{\lambda_1} \ldots x_n^{\lambda_n}).$$

For example, $m_{211}(x_1, x_2, x_3) = x_1x_2x_3(x_1 + x_2 + x_3)$. Also, define the Schur function

$$s_\lambda = \sum_\tau x_1^{\gamma(\tau)_1} \cdots x_n^{\gamma(\tau)_n},$$

where the sum is taken over semistandard tableaux τ on λ with entries in $\{1, \ldots, n\}$. This is equivalent to taking every semistandard tableau τ, replacing each i with x_i, multiplying the elements of τ, and summing.

It is a fact that the Schur functions are symmetric, i.e., they are invariant under any permutation of variables. We will sketch the proof, for completeness. The idea is that any permutation is a composition of adjacent transpositions, so it's enough to show $s_\lambda(x_1, \ldots, x_i, x_{i+1}, \ldots, x_n) = s_\lambda(x_1, \ldots, x_{i+1}, x_i, \ldots, x_n)$. For this, it's enough to show the number of semistandard tableaux with content $(m_1, \ldots, m_i, m_{i+1}, \ldots, m_n)$ is equal to the number of semistandard tableaux with content $(m_1, \ldots, m_{i+1}, m_i, \ldots, m_n)$ for all $(m_1, \ldots, m_n)$. To show this, there's a bijection.

Definition 2.1. For each k, the Bender–Knuth involution σ_k is an involution on the set of semistandard tableaux fixing shape and content except for swapping the number of occurrences of k and $k + 1$. Specifically, the Bender–Knuth involution σ_k is the composition of the following three operations.

1. Simultaneously replace every k with a $k + 1$ and replace every $k + 1$ with a k.
2. If there is a cell containing $k + 1$ above a cell containing k, swap them. This can happen at most once in each column.
3. Sort the cells containing k and $k + 1$ in each row in increasing order.

This implies

$$s_\lambda = \sum_\mu K_{\lambda\mu} m_\mu,$$

where the sum is taken over all partitions μ.

Let ρ be the partition $(n - 1, \ldots, 0)$. We are finally ready to state the formula.

Proposition 2.2 (Jacobi bialternant formula)
For all partitions λ,

$$a_{\lambda+\rho} = a_\rho s_\lambda.$$

Observe that a_ρ is the Vandermonde determinant $\prod_{i<j}(x_i - x_j)$. This implies

$$a_{\lambda+\rho} = a_\rho s_\lambda = \prod_{i<j}(x_i - x_j)\left(\sum_\mu K_{\lambda\mu} m_\mu\right).$$

An example is in order. The most useful case when complex bashing is $n = 3$. When computing symmetric determinants, you will often see expressions such as

$$\begin{vmatrix} 1 & yz(3x^2 - y^2 - z^2) & x(y^2 + z^2) + yz(y + z) \\ 1 & zx(3y^2 - z^2 - x^2) & y(z^2 + x^2) + zx(z + x) \\ 1 & xy(3z^2 - x^2 - y^2) & z(x^2 + y^2) + xy(x + y) \end{vmatrix}.$$

We will evaluate this.

This is

$$\sum_{\text{cyc}} yz(3x^2 - y^2 - z^2)((y(z^2 + x^2) + zx(z + x)) - (z(x^2 + y^2) + xy(x + y))),$$

where the $\sum_{\text{cyc}}$ means to take the sum over the three possible cyclic variants. For example,

$$\sum_{\text{cyc}} x^2 y = x^2 y + y^2 z + z^2 x.$$

Expanding, this becomes

$$\sum_{\text{cyc}} -3x^3y^3z + xy^5z - 3x^2y^3z^2 + y^5z^2 + 3x^3yz^3 + 3x^2y^2z^3 - y^4z^3 + y^3z^4 - xyz^5 - y^2z^5 = \sum_{\text{cyc}} y^5z^2 - y^2z^5 + y^3z^4 - y^4z^3.$$

Writing as a sum of permutations, we get

$$\sum_{w \in S_3} \operatorname{sgn}(w) w(x^5y^2) - \sum_{w \in S_3} \operatorname{sgn}(w) w(x^4y^3) = a_{520} - a_{430}.$$

So by the Jacobi bialternant formula, this becomes

$$a_{210}(s_{310} - s_{220}) = (x - y)(x - z)(y - z)(s_{31} - s_{22}).$$

We will now write the s_λ in terms of m_λ. We only need to sum over partitions μ with sum 4, since otherwise $K_{\lambda\mu} = 0$. We compute the eight Kostka numbers

$$\begin{array}{llll} K_{31,4} = 0 & K_{31,31} = 1 & K_{31,22} = 1 & K_{31,211} = 2 \\ K_{22,4} = 0 & K_{22,31} = 0 & K_{22,22} = 1 & K_{22,211} = 1. \end{array}$$

We do not need to compute $K_{31,1111}$ or $K_{22,1111}$ since $n = 3$ and hence we are only summing over semistandard tableaux with entries in $\{1, 2, 3\}$, so we only need to consider tableaux with at most 3 distinct entries. Hence

$$\begin{aligned} s_{31} - s_{22} &= m_{31} + m_{22} + 2m_{211} - m_{22} - m_{211} \\ &= m_{31} + m_{211} \\ &= x^3y + x^3z + y^3z + y^3x + z^3x + z^3y + xyz(x + y + z) \\ &= (x^2 + y^2 + z^2)(xy + yz + zx). \end{aligned}$$

So our original determinant has the factored form

$$(x - y)(x - z)(y - z)(x^2 + y^2 + z^2)(xy + yz + zx).$$

Bang.

§3 Warm-ups

These problems are relatively straightforward and should be tried before the walkthroughs. Many of these are taken from the OTIS unit `BGW-complex`.

Problem 3.1. In triangle ABC with circumcenter O, point X is the reflection of O across $\overline{BC}$. Define Y, Z similarly. Prove that AX, BY, CZ are concurrent.

Problem 3.2 (ToT Spring 2018/J-A4)**.** Let ABC be a triangle with circumcenter O and let H be the foot of the A-altitude. Let P be the foot of the perpendicular from A to line CO. Prove that line HP bisects side AB.

Problem 3.3 (PUMaC 2016/Finals A3)**.** On a cyclic quadrilateral $ABCD$, let M and N denote the midpoints of $\overline{AB}$ and $\overline{CD}$. Let E be the projection of C onto $\overline{AB}$ and let F be the reflection of N over the midpoint of $\overline{DE}$. Assume F lies in the interior of quadrilateral $ABCD$. Prove that $\angle BMF = \angle CBD$.

Problem 3.4 (PUMaC 2013)**.** Let γ and I be the incircle and incenter of triangle ABC. Let D, E, F be the tangency points of γ to $\overline{BC}$, $\overline{CA}$, $\overline{AB}$ and let D' be the reflection of D about I. Assume EF intersects the tangents to γ at D and D' at points P and Q. Show that $\angle DAD' + \angle PIQ = 180°$.

Problem 3.5 (China 1996)**.** Triangle ABC has orthocenter H. The tangents from A to the circle with diameter $\overline{BC}$ are AP and AQ. Prove that H, P, Q are collinear.

Problem 3.6 (JMO 2025/5)**.** Let H be the orthocenter of acute triangle ABC, let F be the foot of the altitude from C to AB, and let P be the reflection of H across BC. Suppose that the circumcircle of triangle AFP intersects line BC at two distinct points X and Y. Prove that C is the midpoint of XY.

Problem 3.7 (Taiwan TST 2014)**.** In $\triangle ABC$ with incenter I, the incircle is tangent to $\overline{CA}$, $\overline{AB}$ at E, F. The reflections of E, F across I are G, H. Let Q be the intersection of $\overline{GH}$ and $\overline{BC}$, and let M be the midpoint of $\overline{BC}$. Prove that $\overline{IQ}$ and $\overline{IM}$ are perpendicular.

Problem 3.8 (IMO 2024/4)**.** Let ABC be a triangle with $AB < AC < BC$. Let the incenter and incircle of triangle ABC be I and ω, respectively. Let X be the point on line BC different from C such that the line through X parallel to AC is tangent to ω. Similarly, let Y be the point on line BC different from B such that the line through Y parallel to AB is tangent to ω. Let AI intersect the circumcircle of triangle ABC at $P \neq A$. Let K and L be the midpoints of AC and AB, respectively.

Prove that $\angle KIL + \angle YPX = 180°$.

Problem 3.9 (ISL 2024 G1)**.** Let $ABCD$ be a cyclic quadrilateral such that $AC < BD < AD$ and $\angle DBA < 90°$. Point E lies on the line through D parallel to AB such that E and C lie on opposite sides of line AD, and $AC = DE$. Point F lies on the line through A parallel to CD such that F and C lie on opposite sides of line AD, and $BD = AF$.

Prove that the perpendicular bisectors of segments BC and EF intersect on the circumcircle of $ABCD$.

Problem 3.10 (ISL 2018 G7)**.** Let O be the circumcenter and Ω be the circumcircle of an acute triangle ABC. Let P be a point on Ω different from A, B, C, as well as their antipodes. Let O_A, O_B, O_C be the circumcenters of $\triangle AOP$, $\triangle BOP$, $\triangle COP$, respectively. Line ℓ_A passes through O_A and is perpendicular to $\overline{BC}$; lines ℓ_B and ℓ_C are

defined similarly.

Prove that the circumcircle of the triangle determined by ℓ_A, ℓ_B, ℓ_C is tangent to line OP.

Problem 3.11 (USAMO 2023/1)**.** In an acute triangle ABC, let M be the midpoint of $\overline{BC}$. Let P be the foot of the perpendicular from C to AM. Suppose that the circumcircle of triangle ABP intersects line BC at two distinct points B and Q. Let N be the midpoint of $\overline{AQ}$. Prove that $NB = NC$.

Problem 3.12 (IMO 2023/2)**.** Let ABC be an acute triangle with $AB < AC$. Let Ω be the circumcircle of ABC. Let S be the midpoint of the arc CB of Ω containing A. The perpendicular from A to BC meets BS at D and meets Ω again at $E \neq A$. The line through D parallel to BC meets line BE at L. Denote the circumcircle of triangle BDL by ω. Let ω meet Ω again at $P \neq B$. Prove that the line tangent to ω at P meets line BS on the internal angle bisector of $\angle BAC$.

§4 Walkthroughs

§4.1 Canada 1991/3

Example 4.1 (Canada 1991/3)

Let P be a point inside circle ω. Consider the set of chords of ω that contain P. Prove that their midpoints all lie on a circle.

Walkthrough. Problems you can complex bash in your head are rare, but this is one of them. We will (of course) let ω be the unit circle. Let $\overline{AB}$ be a chord of ω containing P.

One possible approach is to solve for b in terms of a and p. However, this is a terrible idea. The problem is symmetric in A and B, so this approach breaks symmetry. We want to keep everything symmetric, so in some sense, we'd like to work with the variables $\sigma := a + b$ and $\pi := ab$, rather than a and b themselves.

(a) Write down the condition that P lies on $\overline{AB}$.

(b) We need to show $\frac{1}{2}(a+b)$ lies on a fixed circle. Use the above condition to rewrite this in terms of ab.

(c) Use $|ab| = 1$ to show that $\frac{1}{2}(p + ab\overline{p})$ lies on a fixed circle.

§4.2 EMC 2024/J3

Example 4.2 (EMC 2024/J3)

Let ABC be a triangle with incenter I and incircle ω. Let ℓ be the tangent to ω parallel to BC and distinct from BC. Let D be the intersection of ℓ and AC, and let M be the midpoint of $\overline{ID}$. Prove that $\angle AMD = \angle DBC$.

Walkthrough. This is the first problem I solved using complex numbers in-contest.

Let ω be the unit circle, and let BC, CA, and AB be tangent to ω at X, Y, and Z, respectively.

(a) Compute a, d, m, b, and c using ice cream cone.

(b) We need to show

$$\frac{a-m}{d-m} \div \frac{d-b}{c-b} \in \mathbb{R}.$$

Show that since $\frac{c-b}{x} \in i\mathbb{R}$, it's equivalent to show

$$\frac{(a-m)x}{(d-m)(d-b)} \in i\mathbb{R}.$$

(c) Finish by writing the above in terms of x, y, and z.

I did not see the optimization of using $\frac{c-b}{x} \in i\mathbb{R}$ during the contest.

§4.3 HMMT Team 2026/6

Example 4.3 (HMMT Team 2026/6)

Let $ABCD$ be a cyclic quadrilateral with circumcenter O, and let M be the midpoint of $\overline{AB}$. Suppose that $\angle CMD = 90^\circ$ and $\overline{AD} \perp \overline{BC}$. Prove that the reflection of O over $\overline{AB}$ lies on line CD.

Walkthrough. This problem took me 15 minutes in-contest. This is a classic example of the power of setting good variables.

Let $(ABCD)$ be the unit circle. We begin by writing the conditions and conclusion in terms of complex numbers.

(a) Show that the conditions are equivalent to

$$ad + bc = 0 \quad \text{and} \quad \frac{(2c - (a+b))(2c - cd(\frac{1}{a} + \frac{1}{b}))}{(2d - (a+b))(2d - cd(\frac{1}{a} + \frac{1}{b}))} = -1.$$

(b) Show that the conclusion is equivalent to $a + b + cd(\frac{1}{a} + \frac{1}{b}) = c + d$.

The second condition and conclusion suggest the substitution $x = a+b$ and $y = cd(\frac{1}{a} + \frac{1}{b})$. Then we need to show the two conditions

$$ad + bc = 0 \tag{★}$$

$$\frac{(2c - x)(2c - y)}{(2d - x)(2d - y)} = -1 \tag{★★}$$

imply $x + y = c + d$.

Usually, you want to use points on the unit circle as variables. However, all we have left now are polynomial equalities (no conjugation). Also, (★★) and the conclusion are far more easily written in terms of x and y than a and b. Therefore, we'll try get rid of a and b completely. The first condition is not symmetric in a and b, though, so we'll replace it with a symmetric condition.

Since the problem is true, it is possible to do some algebra to show (★) and (★★) implies the conclusion. Swapping a and b and doing the same algebra, one can show $ac + bd = 0$ and (★★) implies the conclusion as well. Therefore, we can safely replace (★) with the weaker (but symmetric) condition

$$(ad + bc)(ac + bd) = 0. \tag{†}$$

We can finish now.

(c) Rewrite (†) as $xy + (c - d)^2 = 0$.

(d) Rewrite (★★) as

$$2(c^2 + d^2) - (x + y)(c + d) + xy = 0$$

and substitute in $xy = -(c - d)^2$ to finish.

§4.4 ISL 2023 G5

Example 4.4 (ISL 2023 G5)

Let ABC be an acute triangle with circumcircle ω and circumcenter O. Points $D \neq B$ and $E \neq C$ lie on ω such that $BD \perp AC$ and $CE \perp AB$. Let CO meet AB at X, and BO meet AC at Y.
Prove that the circumcircles of triangles BXD and CYE have an intersection on line AO.

Walkthrough. Let (ABC) be the unit circle.

(a) Compute X and D. Use symmetry to get Y and E.

Since the problem is true, there is some point P such that $P \in \overline{AO} \cap (BXD) \cap (CYE)$. We will solve for p.

(b) Let $\lambda = \frac{p}{a} \in \mathbb{R}$. Use $P \in (BXD)$ and $\lambda \in \mathbb{R}$ to get

$$\frac{c(\lambda b - a)(\lambda c + b)(abc + a^2 b + ac^2 + b^2 c)}{a(\lambda a - b)(\lambda b + c)(abc + a^2 c + ab^2 + bc^2)} = -1.$$

(c) Use $p \in (CYE)$ and $\lambda \in \mathbb{R}$ to obtain a similar condition. This should require no computation. Exploit symmetry.

(d) Multiply the conditions you found in parts **(b)** and **(c)** to obtain

$$\frac{bc(\lambda a - b)(\lambda a - c)}{a^2(\lambda b - a)(\lambda c - a)} = 1.$$

Solve for λ and p. You should get $p = \frac{a^2+bc}{b+c}$.

(e) When we solved for p, we assumed P lies on $\overline{AO}$, (BXD), and (CYE), which is not allowed since this is assuming the problem. Show that $p = \frac{a^2+bc}{b+c}$ actually lies on all three curves to finish.

§4.5 MOP 2025

Example 4.5 (MOP 2025)

Let ω be a circle centered at O, and let Γ be a circle containing ω. Let ℓ be a line tangent to ω. Suppose ℓ meets Γ at P and Q. Show that as ℓ varies, the circumcenter of $\triangle PQO$ lies on a fixed circle.

Walkthrough. I solved this problem on a MOP 2025 team contest. I presented my solution to Carl. After I was done, he said my setup was clean. We went on to win the contest.

Let ω be the unit circle. Let Γ have radius r and be centered at $k \in \mathbb{R}$. Let ℓ be tangent to ω at Z.

(a) Use the fact that p lies on Γ to obtain $p\overline{p} = k(p + \overline{p}) + r^2 - k^2$.

(b) Use the fact that p lies on ℓ to get $\overline{p} = \frac{2z-p}{z^2}$.

(c) By symmetry, repeat the two earlier parts with q instead to get analogous formulas for $q\overline{q}$ and $\overline{q}$.

(d) Since $o = 0$, the circumcenter of $\triangle PQO$ is

$$\begin{vmatrix} p & p\overline{p} \\ q & q\overline{q} \end{vmatrix} \div \begin{vmatrix} p & \overline{p} \\ q & \overline{q} \end{vmatrix}.$$

Substitute in your formulas for $p\overline{p}$, $\overline{p}$, $q\overline{q}$, and $\overline{q}$. Use row operations to simplify. You should get

$$k + \frac{(r^2 - k^2)z}{2}.$$

(e) Conclude that as z varies along a circle, the circumcenter does as well.

§4.6 IMO 2019/6

Example 4.6 (IMO 2019/6)

Let ABC be a triangle with incenter I and incircle ω. Let D, E, F denote the tangency points of ω with $\overline{BC}$, $\overline{CA}$, $\overline{AB}$. The line through D perpendicular to $\overline{EF}$ meets ω again at R (other than D), and line AR meets ω again at P (other than R). Suppose the circumcircles of $\triangle PCE$ and $\triangle PBF$ meet again at Q (other than P). Prove that lines DI and PQ meet on the external $\angle A$-bisector.

Walkthrough. I solved this problem while eating cheese curds at the Minnesota State Fair.

Let X be the intersection of the external $\angle A$-bisector and line DI. We show X lies on line PQ. Let O_1 and O_2 be the centers of (PBF) and (PCE). Let ω be the unit circle.

(a) Show that it's enough to show $XP \perp O_1O_2$.

(b) Compute $r = -\frac{ef}{d}$ and $a = \frac{2ef}{e+f}$. Since A lies on PR, we have $a + pr\overline{a} = p + r$. Solve for p to get

$$p = \frac{ef(2d+e+f)}{2ef+df+de}.$$

(c) Compute

$$x = \frac{4d^2ef}{(e+f)(d^2+ef)}.$$

(d) Obtain

$$o_1 = \frac{ef(2d+e+f)(d-f)}{(d+e)(d+f)(e-f)}$$

by translating (PBF) to the origin.

(e) Convince yourself that we have

$$o_2 = -\frac{ef(2d+e+f)(d-e)}{(d+e)(d+f)(e-f)}$$

by swapping e and f in the expression for o_1. Compute $o_1 - o_2$.

(f) Show $\frac{x-p}{o_1-o_2} \in i\mathbb{R}$ to finish.

§4.7 TST 2017/5

Example 4.7 (TST 2017/5)

Let ABC be a triangle with altitude $\overline{AE}$. The A-excircle touches $\overline{BC}$ at D, and intersects the circumcircle at two points F and G. Prove that one can select points V and N on lines DG and DF such that quadrilateral $EVAN$ is a rhombus.

Walkthrough. Let Ω be the A-excircle. Rename D to X and let Y and Z be the points of tangency of lines AC and AB with Ω, respectively. Let M be the midpoint of AE.

We will complex bash with Ω as the unit circle. For a point $P \neq X$ on Ω, let $\phi(P)$ be the point lying on PX and the perpendicular bisector of AE. We need to show $\phi(f) + \phi(g) = 2m$.

(a) Compute

$$m = \frac{3yz + x(y+z) - x^2}{2(y+z)}.$$

(b) Show that

$$\phi(p) = \frac{p(x^2+yz) - x^2(y+z)}{(y+z)(p-x)}.$$

(c) Let p be either f or g. Use the facts that p, a, b, and c are concyclic and p is unit to get a quadratic with roots f and g. Obtain $f+g$ and fg with Vieta's formulas. You do not need to solve for f and g; in fact, you will not be able to, as this quadratic is irreducible. You should get

$$f+g = \frac{4xyz + (x+y)(y+z)(z+x)}{2(xy+yz+zx)} \qquad fg = \frac{xyz(x+y+z)}{xy+yz+zx}.$$

(d) Using the above expressions for $f+g$ and fg, show that $\phi(f) + \phi(g) = 2m$. (This computation is not short.)

§4.8 USAMO 2024/5

Example 4.8 (USAMO 2024/5)

Point D is selected inside acute $\triangle ABC$ so that $\angle DAC = \angle ACB$ and $\angle BDC = 90^\circ + \angle BAC$. Point E is chosen on ray BD so that $AE = EC$. Let M be the midpoint of BC.

Show that line AB is tangent to the circumcircle of triangle BEM.

Walkthrough. I solved this problem during dinner at Peking Garden after MNYMO Girls.

Let (ABC) be the unit circle. Let T lie on (ABC) such that $\angle TAC = \angle ACB$. Let X be a variable point on (ABC). Redefine D as the intersection between lines BX and AT and E as the intersection between line BX and the perpendicular bisector of AC. Since $\angle TAC = \angle ACB$, we have $\angle DAC = \angle ACB$. We show that if $\angle BDC = 90^\circ + \angle BAC$, then line AB is tangent to (BEM).

(a) Compute $t = \frac{ac}{b}$, $d = \frac{a(x(b^2+bc-ac)-abc)}{b^2x-a^2c}$, and $e = \frac{ac(b+x)}{ac+bx}$.

(b) Show that

$$\angle BDC = 90^\circ + \angle BAC \iff \frac{c-d}{b-d} \div \frac{c-a}{b-a} \in i\mathbb{R} \iff \frac{(c-d)(b-a)}{(b-x)(c-a)} \in i\mathbb{R}.$$

(c) Show that the above is equivalent to

$$b(b-c)x^2 + a(b+c)(b+c-2a)x - a^2c(b-c) = 0. \qquad (\bigstar)$$

(d) Line AB is tangent to (BEM) if and only if $\measuredangle ABE = \measuredangle BME$. Hence we require

$$\frac{a-b}{e-b} \div \frac{b-m}{e-m} = \frac{(a-b)(e-m)}{(x-b)(b-c)} \in \mathbb{R}.$$

Show that this is equivalent to $(\bigstar)$.

§4.9 RMM 2026/5

Example 4.9 (RMM 2026/5)

Let ABC be a triangle with $AB < AC$, let O be its circumcenter and let $XYZT$ be a parallelogram inside triangle ABC such that

$$\angle AXB = \angle AZC, \qquad \angle AZB = \angle AXC,$$
$$\angle AYB = \angle ATC, \qquad \angle ATB = \angle AYC.$$

Prove that the diagonals XZ and YT of the parallelogram intersect on the circumcircle of BOC.

Walkthrough. Complex bash with $\overline{XZ} \cap \overline{YT}$ as the origin, so $x + z = 0$ and $y + t = 0$. Write $p \sim q$ if 0, p, and q are collinear in $\mathbb{C}$, or equivalently, if $q = 0$ or $\frac{p}{q} \in \mathbb{R}$. Write $p \sim q \sim r$ if $p \sim q$, $q \sim r$, and $r \sim p$.

(a) From the first two angle conditions and the fact that $XYZT$ lies inside $\triangle ABC$, obtain

$$(a - x)(a + x) \sim (b - x)(c + x) \sim (b + x)(c - x).$$

Use the fact that $x \sim y$ implies $x \sim y + \lambda x$ for all $\lambda \in \mathbb{R}$ to show that the above implies

$$a^2 - x^2 \sim bc - a^2 \sim x(b - c).$$

(b) Since $b \neq c$, conclude that $x \sim \frac{bc-a^2}{b-c}$, and similarly, $y \sim \frac{bc-a^2}{b-c}$ as well. So $x \sim y \sim \frac{bc-a^2}{b-c}$.

Now we work by cases on whether $\frac{bc-a^2}{b-c} = 0$.

(c) If $\frac{bc-a^2}{b-c} \neq 0$, show that $x \sim y$. Conclude that $XYZT$ is degenerate.

(d) If $\frac{bc-a^2}{b-c} = 0$, we have $\frac{a}{b} = \frac{c}{a}$, so the origin is the center of the spiral similarity sending AB to CA. So it's enough to prove the following:

> Let $\triangle ABC$ be a triangle with circumcenter O. Let D be the center of the spiral similarity sending AB to CA. Then D lies on (BOC).

Complex bash this by setting (ABC) to be the unit circle. (The point D is called the A-Dumpty point.)

§4.10 ISL 2024 G6

Example 4.10 (ISL 2024 G6)

Let ABC be an acute triangle with $AB < AC$, and let Γ be the circumcircle of ABC. Points X and Y lie on Γ so that XY and BC meet on the external angle bisector of $\angle BAC$. Suppose that the tangents to Γ at X and Y intersect at a point T on the same side of BC as A, and that TX and TY intersect BC at U and V, respectively. Let J be the center of the excircle of triangle TUV opposite the vertex T.

Prove that AJ bisects $\angle BAC$.

Walkthrough. I completed the following bash in around 15 minutes after finishing the problems on the math section on the ACT.

Let O be the center of Γ. Let $P \neq A$ be the second intersection of the external bisector of $\angle BAC$ and Γ. Let Q be the intersection of lines AP and BC. Redefine J as the intersection between the internal bisector of $\angle UTV$ and the internal bisector of $\angle BAC$; we show our J is the T-excenter of $\triangle TUV$.

Complex bash with Γ as the unit circle.

(a) Prove that $p^2 = bc$.

(b) Show that the internal bisector of $\angle UTV$ passes through O, hence $\frac{j}{t} \in \mathbb{R}$. Use this along with the fact that $AP \perp AJ$ to compute $j = \frac{xy(a-p)}{xy-ap}$.

Since J and T lie on opposite sides of line UV, J cannot be the incenter of $\triangle TUV$. Hence if we show the distance from J to BC is equal to the distance from J to TX, we are done.

(c) Show $|j + \overline{j}bc - b - c| = |j + \overline{j}x^2 - 2x|$ is equivalent.

(d) Convince yourself it's enough to show

$$\frac{j + \overline{j}bc - b - c}{p} = \frac{j + \overline{j}x^2 - 2x}{x}.$$

(e) Multiply by p and write $j = xy\overline{j}$ to get

$$(xy + bc)\overline{j} - (b + c) = p(x + y)\overline{j} - 2p. \tag{$\bigstar$}$$

At this point, we are happy because this expression is symmetric in x and y.

In some sense, the values a, p, x, y, b, and c that appear in the above equation (a appearing implicitly through our expression of $\overline{j}$) are related to each other "through" the point q, since the lines AP, XY, and BC all pass through Q. In particular, we have the three relations

$$q + \overline{q}bc = b + c$$
$$q + \overline{q}xy = x + y$$
$$q + \overline{q}ap = a + p.$$

So things like $b + c$ and $x + y$ have "nice" expressions in terms of q and $\overline{q}$, but they don't necessarily have nice expressions in terms of each other. So the easiest way to relate these expressions is by writing them in terms of q and $\overline{q}$.

(f) Write (★) in terms of a, p, q, $\overline{q}$, and xy by substituting $bc = p^2$, $b + c = q + \overline{q}p^2$, and $x + y = q + \overline{q}xy$. You should get

$$(xy + p^2)\overline{j} - (q + \overline{q}p^2) = p(q + \overline{q}xy)\overline{j} - 2p.$$

(g) Expand with $\overline{j} = \frac{a-p}{xy-ap}$.

(h) We haven't used the fact that Q lies on line AP yet. Use $q + \overline{q}ap = a + p$ to finish.

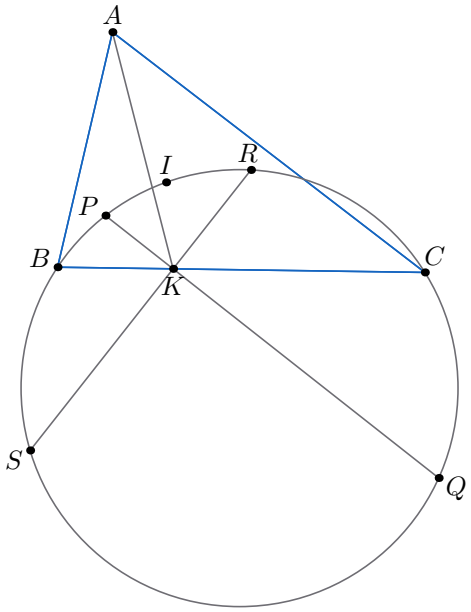
§4.11 ISL 2024 G5

Example 4.11 (ISL 2024 G5)

Let ABC be a triangle with incenter I, and let Ω be the circumcircle of triangle BIC. Let K be a point in the interior of segment BC such that $\angle BAK < \angle KAC$. The angle bisector of $\angle BKA$ intersects Ω at points W and X such that A and W lie on the same side of BC, and the angle bisector of $\angle CKA$ intersects Ω at points Y and Z such that A and Y lie on the same side of BC.
Prove that $\angle WAY = \angle ZAX$.

Walkthrough. At MOP, Razzi (the LeBron of complex bashing) presented a solution to this problem after a MOP test. Revisiting this setup a few months later, I found a further insight to work in terms of pq instead of just p from the beginning. This makes the computation noticeably nicer and (in my opinion) more motivated. Anyway.

We will set Ω as the unit circle. Let j represent I.



(a) Solve for a in terms of j, b, and c. Convince yourself that $a/\overline{a}$ should be j^2, then verify your expression for a satisfies this.

Let P be any point on Ω, and let Q be the other intersection of PK with Ω. Let R and S be the intersections of the line through K perpendicular to PQ with Ω. We show that if P lies on either of the bisectors of $\angle BKA$, then $\angle PAR = \angle QAS$.

(b) Show that it's equivalent to require

$$\frac{(p-k)^2}{(a-k)(b-c)} \in \mathbb{R} \tag{$\dagger$}$$

to imply

$$\frac{(a-p)(a-q)}{(a-r)(a-s)} \in \mathbb{R}. \tag{$\bigstar$}$$

The key insight is to parameterize the entire setup in terms of pq. This is motivated by the following facts:

- There is an inherent symmetry between the points p and q, since they are the roots of some quadratic. Furthermore, this quadratic is likely irreducible, since the condition and conclusion are both symmetric in P and Q.
- Since the quadratic with roots p and q is irreducible, it is easier to work with $p+q$ and pq rather than p and q themselves. In particular, the conditions on p and q are quartic, while the conditions on pq are quadratic.
- Once pq is known, $p+q$ is also known, from the relation $p+q = k+\overline{k}pq$. Additionally, once pq is known, so is rs, since $PQ \perp RS$, which implies $pq + rs = 0$. Similarly, one may find $r+s$.
- Since both the condition and the conclusion are symmetric with respect to swapping p and q or swapping r and s, they are expressible in terms of pq and rs, by (say) the fundamental theorem of elementary symmetric polynomials.

Keeping this in mind, we will rewrite (†) and (★) in terms of pq.

(c) Respecting symmetry, we rewrite (†) as

$$\frac{(p-k)(q-k)}{(a-k)(b-c)} \in \mathbb{R}.$$

Show that this is equivalent to

$$p^2q^2(a-j^2\overline{k}) + bcj^2(a-k) = 0. \tag{††}$$

(d) Show that

$$\frac{(a-p)(a-q)}{(a-r)(a-s)} = \frac{a^2-ak+(1-a\overline{k})pq}{a^2-ak-(1-a\overline{k})pq}.$$

(e) Show that (★) rearranges to

$$p^2q^2(1-a\overline{k})(a-j^2\overline{k}) + j^2(k-a)(j^2-ak) = 0. \tag{★★}$$

(f) Finish by showing (††) and (★★) are equivalent.

§4.12 IMO 2011/6

Example 4.12 (IMO 2011/6)

Let ABC be an acute triangle with circumcircle Γ. Let ℓ be a tangent line to Γ, and let ℓ_a, ℓ_b, and ℓ_c be the lines obtained by reflecting ℓ in the lines BC, CA, and AB, respectively. Show that the circumcircle of the triangle determined by the lines ℓ_a, ℓ_b, and ℓ_c is tangent to the circle Γ.

Walkthrough. Evan has a 15-minute computation that requires guessing a homothety; however, this is black magic to me. We present three solutions that all require little synthetic insight. The solutions are of increasing order of instructiveness.

Let P be the tangency point of ℓ to (ABC). Let $X = \ell_b \cap \ell_c$ and define Y and Z similarly. We complex bash with (ABC) as the unit circle. The first step is the same in all three solutions.

(a) Compute

$$x = \frac{p^2(ab+bc+ca) - 2pabc + a^2bc}{p^2(b+c)} = a + \frac{bc(p-a)^2}{p^2(b+c)} \quad \text{and} \quad \overline{x} = \frac{1}{a} + \frac{(p-a)^2}{a^2(b+c)}.$$

We proceed with the first solution, which I did on Twitch. The computation took two hours, but I had emotional support in the form of chat.

(b) The condition that a point T lies on (XYZ) may be written as

$$\frac{(T-y)(x-z)}{(T-z)(x-y)} \in \mathbb{R}.$$

Compute $\frac{x-z}{x-y}$. It should be really nice.

(c) Substituting in $\overline{T} = \frac{1}{T}$ yields a quadratic whose roots are the intersection of (ABC) and (XYZ); we need to show this quadratic has a double root. Show that this quadratic is

$$T^2(b^2\overline{y} - c^2\overline{z}) - T(b^2 - c^2 + b^2\overline{y}z + c^2y\overline{z}) + b^2z - c^2y = 0.$$

(d) Substitute to show that the discriminant of this quadratic is zero. (This computation is not short.)

The second solution is by `malcolm` and makes use of Casey's theorem.

Theorem (Casey's theorem)

Four circles c_1, c_2, c_3, and c_4 are tangent to a fifth circle if and only if for some choice of signs,

$$T_{12}T_{34} \pm T_{13}T_{42} \pm T_{14}T_{23} = 0$$

where T_{ij} is the length of a common tangent to circles c_i and c_j.

In the case where the radii of c_i for $1 \le i \le 4$ are zero, this specializes to Ptolemy's theorem. Let Γ be (ABC) and let Γ_X, Γ_Y, and Γ_Z be circles of radius zero centered at X, Y, and Z. Then the problem requires Γ, Γ_X, Γ_Y, and Γ_Z to all be tangent to (XYZ). By Casey's theorem, this is equivalent to

$$\sqrt{\operatorname{Pow}_\Gamma(X)}YZ \pm \sqrt{\operatorname{Pow}_\Gamma(Y)}ZX \pm \sqrt{\operatorname{Pow}_\Gamma(Z)}XY = 0.$$

(e) Compute

$$\operatorname{Pow}_\Gamma(X) = x\overline{x} - 1 = \frac{(p-a)^2((ab+bc+ca)p^2 - 2abcp + abc(a+b+c))}{a^2p^2(b+c)^2}$$
$$y - z = \frac{(b-c)((ab+bc+ca)p^2 - 2pabc + abc(a+b+c))}{p^2(a+b)(a+c)}.$$

(f) Divide by unit and symmetric terms to obtain

$$|a-p||b-c| \pm |b-p||c-a| \pm |c-p||a-b| = 0.$$

Conclude by Ptolemy's theorem.

Why should we have expected to use Ptolemy's theorem a priori?

The third solution is by `pieater314159`. We will need the following fact.

Theorem (Möbius transformations send clines to clines)

A cline is either a circle or a line.[a] A Möbius transformation is one of the form $z \mapsto \frac{az+b}{cz+d}$ where $a, b, c, d \in \mathbb{C}$ and $ad - bc \neq 0$. Then

- Möbius transformations send clines to clines, and
- given complex numbers A, B, C, X, Y, and Z such that A, B, and C are pairwise distinct and X, Y, and Z are pairwise distinct, there's exactly one Möbius transformation sending A to X, B to Y, and C to Z.

[a] A line is a circle passing through the point at infinity.

(g) Show that the map

$$f : t \mapsto \frac{p^2(ab+bc+ca) - 2pabc + abct}{p^2(a+b+c-t)}$$

is a Möbius transformation sending (ABC) to (XYZ).

Don't do any computations yet! We need to show $|f(t)| = 1$ for exactly one value of t on the unit circle. If $f(t) = \frac{At+B}{Ct+D}$, then this is equivalent to $|-\frac{B}{A} - t| = |-\frac{D}{C} - t|$. This is the perpendicular bisector of the points $-\frac{B}{A}$ and $-\frac{D}{C}$, which is a line. We need to show this line is tangent to the unit circle.

(h) We need to show the line

$$\left|2p - p^2\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) - t\right| = |a+b+c-t|$$

is tangent to the unit circle. Show that this line is just ℓ. Conclude.

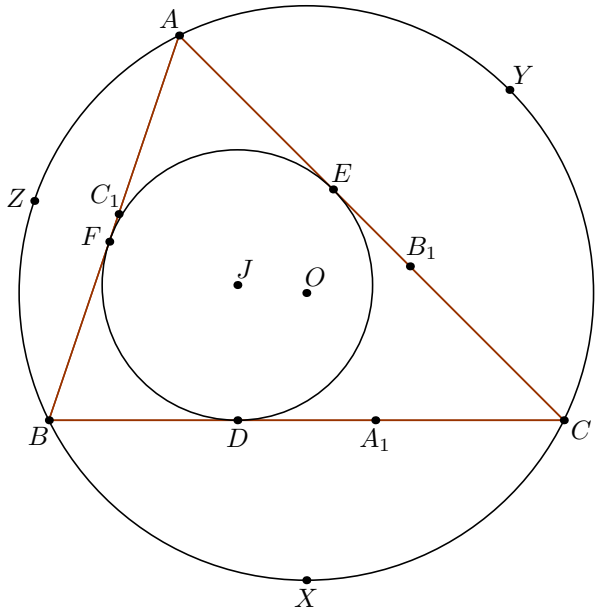
§4.13 IMO 2013/3

Example 4.13 (IMO 2013/3)

Let the excircle of triangle ABC opposite the vertex A be tangent to the side BC at the point A_1. Define the points B_1 on CA and C_1 on AB analogously, using the excircles opposite B and C, respectively. Suppose that the circumcenter of triangle $A_1B_1C_1$ lies on the circumcircle of triangle ABC. Prove that triangle ABC is right.

Walkthrough. This one's a tough cookie. Inspired by `GeronimoStilton`'s famous 2021 computer-assisted complex bash for this problem, I sought to find a setup acceptable for humans. The following bash is the result. See my post on the AoPS thread for a full solution.

Let O be the circumcenter of $\triangle ABC$. Let J and r be the incenter and inradius of $\triangle ABC$. Let X be the arc midpoint of $\overset{\frown}{BC}$ opposite A. Let D be the touchpoint of the incircle of $\triangle ABC$ to $\overline{BC}$. Define Y, Z, E, and F similarly.



First, make sure you've read Section 2.3. We will employ American Smartass Notation: let $|f, g, h|$ for rational functions f, g, and h denote the magnitude of the determinant of the 3×3 matrix whose rows are the three cyclic variations of (f, g, h). For example,

$$\left|a^2+d^2, \frac{a(x\bar{j}+1)}{a+d}, 1\right| = \left\| \begin{vmatrix} a^2+d^2 & \frac{a(x\bar{j}+1)}{a+d} & 1 \\ b^2+e^2 & \frac{b(y\bar{j}+1)}{b+e} & 1 \\ c^2+f^2 & \frac{c(z\bar{j}+1)}{c+f} & 1 \end{vmatrix} \right\|,$$

where the outside vertical bars denote the magnitude.

Complex bash by setting (ABC) as the unit circle. The condition that the circumcenter of $(A_1B_1C_1)$ has magnitude 1 is equivalent to

$$1 = \frac{|1, a_1, a_1\overline{a_1}|}{|1, a_1, \overline{a_1}|}.$$

We will simplify this as much as possible. Throughout, use elementary column operations as needed to assist simplification.

(a) Since $\overline{BD} = \overline{CA_1}$, we have $a_1 + d = b + c$. Also, since D lies on $\overline{BC}$, we have $d + bc\overline{d} = b + c$. So $a_1 = bc\overline{d}$. Use this to show

$$1 = \frac{|1, bc\overline{d}, d\overline{d}|}{|1, bc\overline{d}, ad|}.$$

(b) Observe that $d = j + rx$, so $d\overline{d} = j\overline{j} + r^2 + r(\frac{j}{x} + x\overline{j})$. Also, $d = \frac{1}{2}(j + b + c - bc\overline{j})$, so $bc\overline{d} = \frac{1}{2}(bc\overline{j} + b + c - j)$. Use this to show

$$1 = \frac{2r|1, bc\overline{j} - a, \frac{j}{x} + x\overline{j}|}{|1, bc\overline{j} - a, a(j - \frac{1}{j})|}.$$

(c) By Euler's theorem, $OI^2 = R(R - 2r) = 1 - 2r$, hence $2r = 1 - j\overline{j}$. Use this to show

$$1 = \frac{|1, bc\overline{j} - a, \frac{j}{x} + x\overline{j}|}{|1, bc, a|}.$$

(d) Use $j = x + y + z$ to show

$$1 = \frac{|1, bc\overline{j} - a, \frac{z}{y} + \frac{y}{z}|}{|1, a, bc|}.$$

Now pick p, q, and r such that $a = p^2$, $b = q^2$, $c = r^2$, $x = -qr$, $y = -rp$, and $z = -pq$ (so forget the definition of r as the inradius). We will let the inputs to a_λ, s_λ, and m_λ always be p, q, and r.

(e) Obtain

$$1 = \frac{|1, qr(p^3 + pqr + q^2r + qr^2), p(q^2 + r^2)|}{|1, p^2, q^2r^2|}.$$

(f) If $p^2 = q^2$, then the determinant in the denominator is zero; this implies $p^2 - q^2$ divides it. Similarly, $p^2 - r^2$ and $q^2 - r^2$ divide it. Yet it is a degree-3 polynomial in p^2, q^2, and r^2 (so p^2, q^2, and r^2 are "degree-1"). Match the coefficients of the p^4q^2 term to conclude the denominator is $|(p^2 - q^2)(p^2 - r^2)(q^2 - r^2)|$.

(g) The determinant in the numerator is

$$\sum_{\text{cyc}} qr(p^3 + pqr + q^2r + qr^2)(q(r^2 + p^2) - r(p^2 + q^2)).$$

Expand to show it is $a_{521} - a_{431} - a_{530}$. This should not be hard: although there are sixteen terms, any of the form $p^aq^br^c$ where a, b, and c are not all distinct should cancel, so you can just ignore them. (Why?)

(h) Apply the Jacobi bialternant formula to obtain

$$\left|\frac{s_{311} - s_{221} - s_{32}}{(p + q)(q + r)(r + p)}\right|.$$

(i) Expand in terms of m_λ to obtain

$$1 = \left|\frac{m_{32} + 2m_{221}}{m_{21} + 2m_{111}}\right|.$$

Square to get

$$1 = \frac{(m_{32} + 2m_{221})(m_{2,-1} + 2m_1)}{(m_{21} + 2m_{111})^2},$$

then expand to get

$$m_{32}m_{2,0,-1} + 2m_{221}m_{2,0,-1} + 2m_{32}m_1 + 4m_{221}m_1 - m_{21}^2 - 4m_{21}m_{111} - 4m_{111}^2 = 0.$$

We're going to expand this. The key insight is that writing $m_\mu m_\nu$ as a sum of m_λ isn't so bad, since you can abuse symmetry. For example, suppose we want to compute

$$m_{32}m_{2,-1} = (x^3y^2+x^3z^2+y^3x^2+y^3z^2+z^3x^2+z^3y^2)\left(\frac{x^2}{y}+\frac{x^2}{z}+\frac{y^2}{x}+\frac{y^2}{z}+\frac{z^2}{x}+\frac{z^2}{y}\right).$$

A full expansion proceeds by multiplying each monomial in the left term with the right term. However, the terms on the left side are "all essentially the same," the only difference being a permutation of variables. So the monomial x^3y^2 multiplied by the right term is

$$x^5y + \frac{x^5y^2}{z} + x^2y^4 + \frac{x^3y^4}{z} + x^2y^2z^2 + x^3yz^2.$$

Since the left side is symmetric, we get all symmetric variants of these terms when doing the full expansion. So the full expansion must be

$$m_{51} + m_{5,2,-1} + m_{42} + m_{4,3,-1} + 6m_{222} + m_{321}.$$

In this way, we obtain the expansion of $m_{32}m_{2,-1}$ by doing only 6 multiplications instead of 36.

(j) Expand all $m_\nu m_\mu$ as above to get

$$m_{63} + m_{54} + m_{621} + 3m_{432} + 4m_{531} + 2m_{441} + 2m_{522} + 8m_{333} = 0.$$

(k) The condition that $\triangle ABC$ is right may be encoded as $(a+b)(b+c)(c+a) = 0$, or $(p^2+q^2)(q^2+r^2)(r^2+p^2) = 0$. Hence we know $(p^2+q^2)(q^2+r^2)(r^2+p^2) = m_{42}+2m_{222}$ divides the left-hand side of the above; we will find the other degree-3 factor. To this end, compute the product of $m_{42} + 2m_{222}$ with every degree-3 monomial symmetric function. You should get

$$\begin{aligned}(m_{42} + 2m_{222})m_{21} &= m_{63} + m_{621} + m_{54} + 2m_{522} + 2m_{441} + 3m_{432}\\ (m_{42} + 2m_{222})m_3 &= m_{72} + m_{54} + m_{432} + 2m_{522}\\ (m_{42} + 2m_{222})m_{111} &= m_{531} + 2m_{333}.\end{aligned}$$

(l) Factor using the above to get

$$(m_{42} + 2m_{222})(m_{21} + 4m_{111}) = 0.$$

(m) To finish, it's enough to show $m_{21} + 4m_{111} \neq 0$, or

$$\frac{p}{q} + \frac{q}{p} + \frac{q}{r} + \frac{r}{q} + \frac{r}{p} + \frac{p}{r} + 4 \neq 0.$$

Show that this is equivalent to

$$\cos(\angle POQ) + \cos(\angle QOR) + \cos(\angle ROP) = -2.$$

(n) Show that

$$\cos(\angle POQ) + \cos(\angle QOR) + \cos(\angle ROP) \geq -\frac{3}{2}$$

by Jensen's inequality to finish.